DWSIM Simulation of the Benzoic Acid Plant

Cross-check of the Group 22 mass and energy balances

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Summary

The full benzoic acid process, including the toluene recycle, was simulated in DWSIM and converged. The simulation confirms the group's material balance around the reactor, flash drum and vent to within 1–5%, producing 15.59 kmol/h (1,904 kg/h) of benzoic acid from 17 kmol/h of fresh toluene, about 9% more than the hand-calculated 14.33 kmol/h.

The energy balance agrees less well. The simulated reactor cooling duty is 1,449 kW against 3,565 kW by hand; the column duties are about four times the hand values. Both differences trace to specific assumptions in the hand calculations, set out in Section 6, rather than to the simulation.

- **Reactor:** 15% per-pass conversion of 103.95kmol/h toluene gives 15.59 kmol/h benzoic acid; O_2, toluene and water balances close exactly.
- **Vent and flash drum:** vent gas 102.8 vs 105.1 kmol/h by hand; condensate 63.1 vs 61.6 kmol/h.
- **Water:** the simulation removes all 15.4 kmol/h of reaction water at the decanter; the hand balance removes only 7.6 kmol/h, so water would build up in the recycle.
- **Overall closure:** mass in 5,172.6 kg/h, out 5,136.9 kg/h (0.69% gap), entirely explained by the recycle convergence tolerance.
- **Production:** 15,234 t/yr at 8,000 h/yr, so the fresh feed can be scaled by 0.788 to meet the 12,000 t/yr target.
- **Safety:** reactor off-gas O_2/N_2 ratio is 0.029, well inside the 0.0625 flammability limit used in the reactor design.

1. Aim and scope

The group designed the plant by hand: mass and energy balances (Report 1), reactor sizing (Report 2), heat exchangers (Report 3) and vessels, pumps and compressors (Report 4). These calculations use simplified vapour–liquid equilibrium, iterate the recycle only twice and estimate enthalpies by hand. The aim of this work was to rebuild the same process in DWSIM, on the same basis and assumptions and test whether the hand results are internally consistent.

The simulation uses the hand-calculation basis throughout: about 100 kmol/h toluene into the reactor, 125 kmol/h air (79% N_2, 21% O_2), 15% toluene conversion per pass and a reactor at 165°C and 8 bar. Results are therefore directly comparable before scaling to 12,000 tonne/yr.

2. Simulation method

Raoult's law was used for every unit, matching the ideal vapour–liquid equilibrium assumed in Report 1. The decanter alone uses a copy of the package with immiscible water enabled, because an ideal model cannot otherwise form two liquid phases.

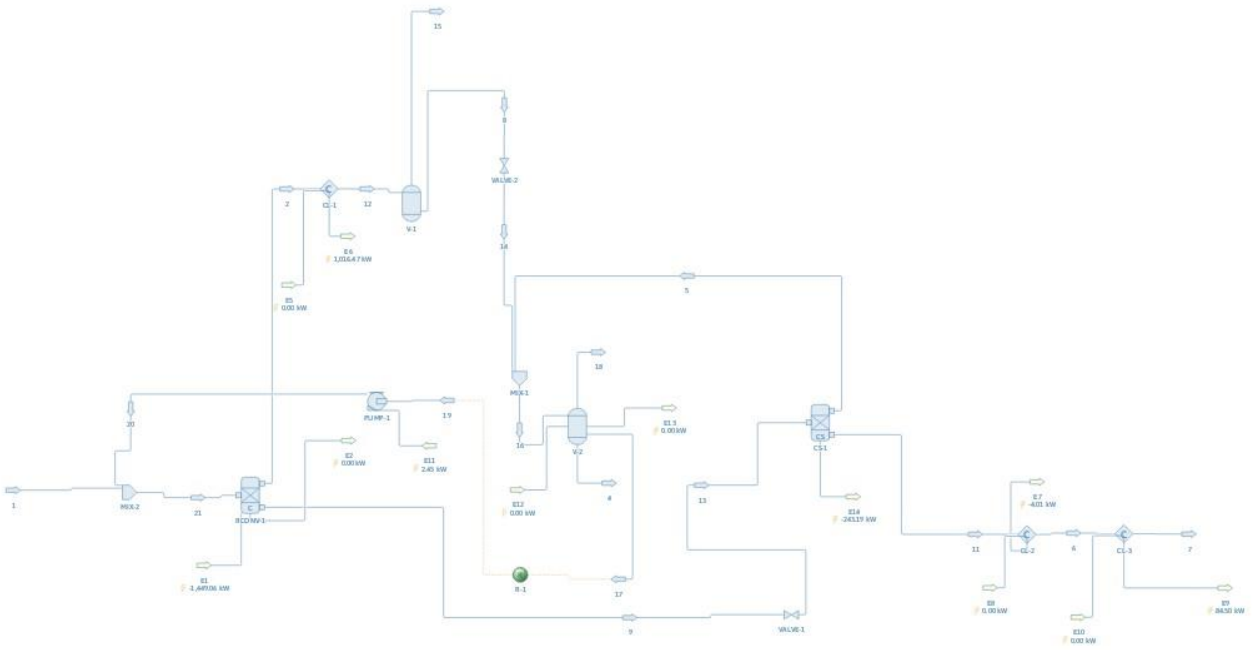


Figure 1. DWSIM flowsheet. Stream 1 is fresh feed; streams 17 → 19 → 20 are the toluene recycle through R-1 and PUMP-1; stream 7 is the product. Energy streams (E-numbers) carry the unit duties.

Unit (PFD)	DWSIM block	Specification
Reactor	RCONV-1 (conversion reactor)	$2C_7H_8 + 3O_2 \rightarrow 2\ C_7H_6O_2 + 2\ H_2O$; 15% toluene conversion; 165°C, 8 bar
Cooler-condenser	CL-1	Outlet 45°C, no pressure drop
Flash drum	V-1	Flash at 45°C, 8 bar; N_2/O_2 vented (stream 15)
Letdown valves	VALVE-1, VALVE-2	Outlet 1.01325 bar
Decanter	V-2 (three-phase separator)	Water to stream 4; organic layer to recycle (stream 17)
Distillation column	CS-1 (compound separator)	99% of benzoic acid to bottoms; all other components overhead
Product coolers	CL-2, CL-3	Product cooled through its melting point (solidification)
Toluene recycle	R-1, PUMP-1, MIX-2	Decanter organics pumped to 8 bar and mixed with fresh feed

Table 1. Unit operations and key specifications.

Simplifications

Three simplifications were needed to reach a converged flowsheet. Each one limits what the simulation can confirm.

- **Column as a component split.** A rigorous column (Wang–Henke, then Napthali–Sandholm) failed to converge. A shortcut column (reflux ratio 1.5, total condenser, 1 atm) gave duties but no product flows, so the separation was set as 99% benzoic acid recovery. Product purity is therefore an input, not a result.
- **Separator outlet temperature.** The compound separator returns both products at the feed temperature (125.3°C) rather than 110°C top and 250°C bottom. The simulated bottoms-cooler duty is therefore not comparable with the hand value.
- **Feed pressure.** Fresh air and toluene enter at 8 bar, so the air compressors of Report 4 are not modelled.

3. Simulation results

The flowsheet and recycle converged (DWSIM average recycle error 0.17%). Table 2 gives the component flows of the key streams; the full set is in the Appendix.

Stream	Description	T (°C)	Toluene	O_2	N_2	Benzoic acid	Water	Total
1	Fresh feed	25.0	17.00	26.25	98.75	0	0	142.00
21	Reactor inlet	34.8	103.95	26.25	98.75	0.54	0	229.49
2	Reactor vapour	165.0	52.70	2.86	98.75	0.38	11.27	165.96
9	Reactor liquid	165.0	35.66	~0	~0	15.75	4.33	55.73
15	Vent gas	45.0	1.02	2.86	98.75	~0	0.22	102.84
8	Flash-drum liquid	45.0	51.68	~0	~0	0.38	11.05	63.12
4	Decanter water	40.0	~0	~0	~0	0	15.38	15.38
5	Column overhead	125.3	35.66	~0	~0	0.16	4.33	40.14
20	Toluene recycle	40.2	86.95	~0	~0	0.54	0	87.49
7	Product	122.4	0	0	0	15.59	0	15.59

Table 2. Key stream molar flows (kmol/h). Pressures are 8 bar upstream of the letdown valves and 1.01325 bar downstream.

Material balance closure

Every component balance closes around the reactor. Of the 103.95 kmol/h toluene entering, 15.59 kmol/h (15.0%) reacts. This consumes 23.39 kmol/h O_2 (1.5 × 15.59), leaving 2.86 kmol/h in the vent and makes 15.59 kmol/h each of benzoic acid and water, matching the stoichiometry.

Basis	In (kg/h)	Out (kg/h)	Difference
Whole plant: stream 1 in; streams 4, 7, 15, 18 out	5,172.65	5,136.90	35.75 kg/h (0.69%)
Toluene (kmol/h)	17.00 fresh	15.59 reacted + 1.02 vent	0.39 kmol/h
Water (kmol/h)	15.59 formed	15.38 decanter + 0.22 vent	0.00
Benzoic acid (kmol/h)	15.59 formed	15.59 product	0.00
Nitrogen (kmol/h)	98.75	98.75 vent	0.00

Table 3. Overall and component balances.

The only gap is 0.39 kmol/h of toluene, worth 35.8 kg/h. It equals the difference between the recycle streams either side of the tear (87.34 kmol/h in stream 19 against 86.95 kmol/h in stream 20). The 0.69% closure error is therefore the recycle tolerance, not lost material and tightening the R-1 tolerance would remove it.

4. Comparison with the hand mass balance

Figure 2 and Table 4 compare the simulation with the final hand-calculated stream tables in Report 1. Reactor and flash-drum totals agree within 1–5%; the larger differences are in how components split between phases and in where the water leaves.

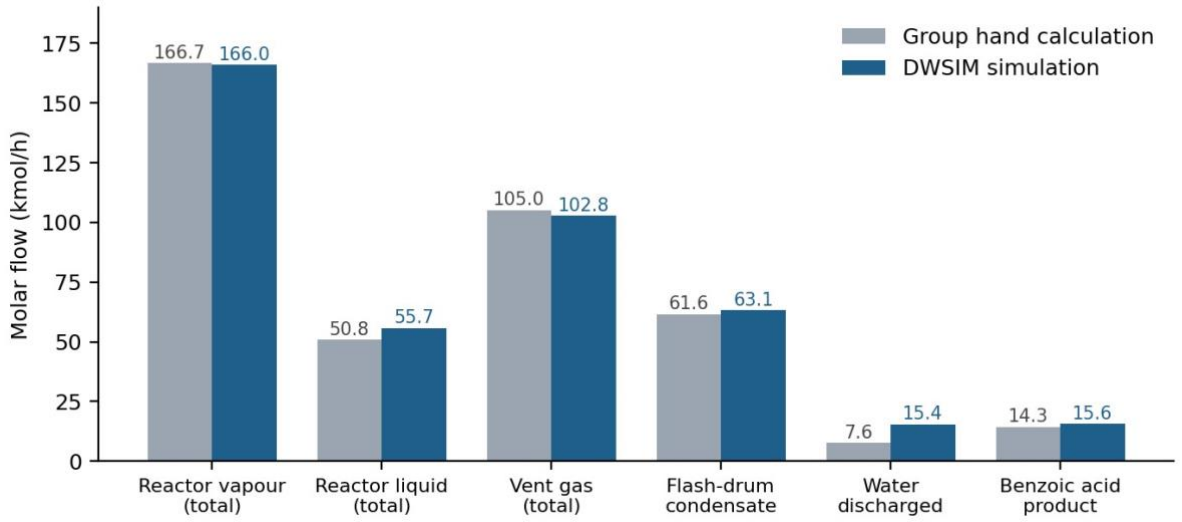


Figure 2. Key stream totals: group hand calculation against DWSIM.

Quantity (kmol/h)	Hand (Report 1)	DWSIM	Difference
Reactor vapour: total	166.67	165.96	−0.4%
Reactor vapour: toluene / water	56.30 / 7.87	52.70 / 11.27	−6% / +43%
Reactor liquid: total	50.83	55.73	+10%
Reactor liquid: toluene / benzoic acid	28.70 / 15.00	35.66 / 15.75	+24% / +5%
Vent gas: total	105.05	102.84	−2.1%
Vent gas: toluene / water	1.297 / 1.257	1.018 / 0.216	−22% / −83%
Vent gas: O_2	3.75	2.86	−24%
Flash-drum condensate	61.61	63.12	+2.4%
Water discharged from decanter	7.61	15.38	+102%
Toluene recycled to reactor	54.00 + 29.40	86.95	+4%
Benzoic acid product	14.33	15.59	+8.8%
Product purity (mol%)	97.5	100 (specified)	–

Table 4. Hand calculation against simulation, same basis.

Reactor phase split

The group found the reactor vapour by fixing the outlet O_2 mole fraction at 0.0225, then applied Raoult's law with vapour pressures at 165°C (382 kPa toluene, 703 kPa water) at 800 kPa. DWSIM solves the same equilibrium rigorously and reaches almost the same total vapour (165.96 against 166.67 kmol/h). It puts more water into the vapour and more toluene into the liquid because it solves the liquid composition and vapour fraction together rather than from a fixed O_2 fraction.

Flash drum and vent

Vent N_2 agrees exactly and total vent flow is within 2%. The vent carries less toluene and much less water in the simulation. The hand method gave each component its full vapour pressure at 45°C (9.88 kPa toluene, 9.57 kPa water), which is the correct treatment for two immiscible liquids. DWSIM's ideal Raoult's law instead treats them as one miscible liquid, so water (17.5 mol% of the condensate) exerts only 17.5% of its vapour pressure. On this point the hand calculation is physically closer to reality; the simulation understates water loss through the vent. Vent O_2 is lower in the simulation (2.86 against 3.75 kmol/h) because 103.95 kmol/h of toluene reaches the reactor rather than 100, so slightly more O_2 is consumed.

Decanter and water balance

This is the most important difference. The reaction forms 15 kmol/h of water. The hand balance removes 1.26 kmol/h in the vent, 7.61 kmol/h at the decanter and 0.06 kmol/h in the product, leaving about 6 kmol/h that returns with the distillate. At steady state that water has nowhere to leave, so it would accumulate around the recycle loop. In the simulation the distillate passes through the decanter, so all reaction water (15.38 kmol/h, 277 kg/h, 99.999% pure) leaves in stream 4 and the recycle carries none. The hand balance should be revised so the decanter removes all net water, which roughly doubles the waste-water flow the decanter must handle.

Recycle and production rate

The group iterated the recycle twice and reached 14.33 kmol/h benzoic acid, noting that more iterations would approach 15 kmol/h. The converged simulation confirms this: with 17 kmol/h fresh toluene the loop settles at 86.95 kmol/h recycled toluene and 103.95 kmol/h into the reactor, giving 15.59 kmol/h product. Only 0.54 kmol/h benzoic acid recycles to the reactor, against 1.04 kmol/h assumed by hand, because the simulated column recovers 99% to the bottoms.

5. Energy balance comparison

Figure 3 and Table 5 compare unit duties. The cooler-condenser agrees within 17%; the reactor, column and solidification duties differ substantially, for the reasons below.

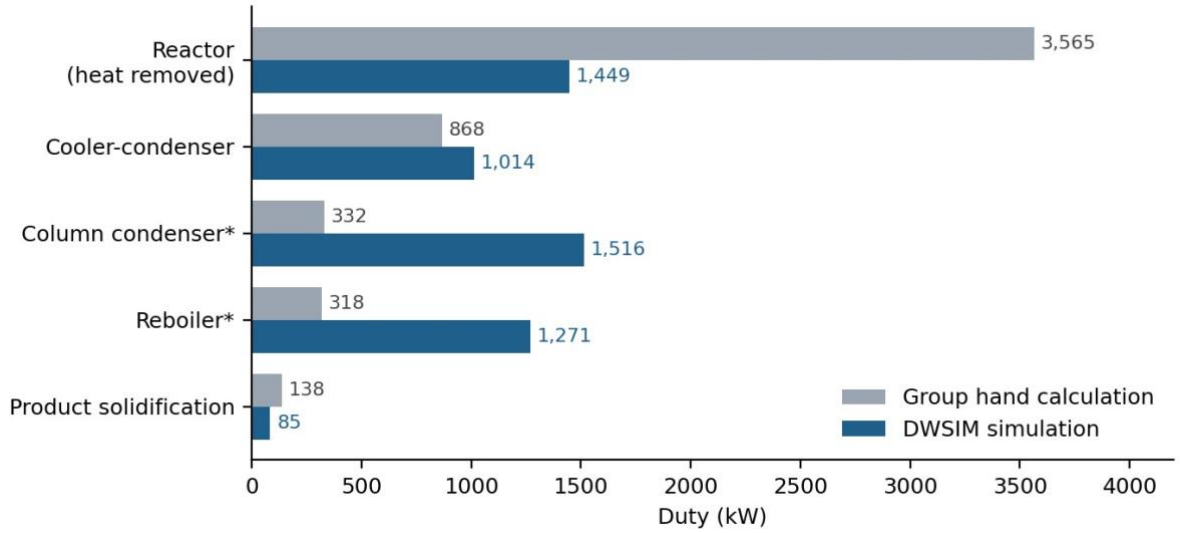


Figure 3. Unit duties, hand calculation (Reports 1 and 3) against DWSIM. *Column duties from the DWSIM shortcut column on the once-through flowsheet at reflux ratio 1.5.

Unit	Hand (kW)	DWSIM (kW)	DWSIM energy stream
Reactor (heat removed)	3,565	1,449	E1
Cooler-condenser CL-1	868	1,014	E6
Column condenser	332	1,516*	Shortcut column
Reboiler	318	1,271*	Shortcut column
Bottoms cooler (250 → 125 °C)	94	not comparable	E7 (4.0 kW heating)
Product solidification	138	85	E9
Recycle pump	–	2.45	E11

Table 5. Unit duties. Hand reactor duty is 12,835,538 kJ/h ÷ 3,600.

Reactor duty

The hand reactor duty of 3,565 kW is larger than the heat the reaction can release. Using the group's own formation enthalpies (toluene +12.0, benzoic acid −290.2, water −286.6 kJ/mol), converting 15 kmol/h of toluene releases at most about 2,450 kW. Some of that heat is then absorbed in heating the feed to 165°C and vaporising about 53 kmol/h of toluene, so the net cooling duty must be well below 2,450 kW. The simulated 1,449 kW is consistent with this.

A likely source of the excess is the reactor inlet enthalpy in Report 1. The fresh toluene is entered as H = 100 × (12,000 + 30,046) kJ, which adds the enthalpy at 25°C and at 165°C together instead of using one inlet state. The same pattern appears for the recycle stream. Recalculating with the feed at its actual inlet temperature should bring the hand duty close to the simulation. If the cooling jacket and 254.5 m coil in Report 2 were sized on this duty, they would be oversized by roughly a factor of 2.5.

Cooler-condenser

The simulated duty is 1,014 kW against 868 kW. DWSIM condenses more water (11.05 against 7.61 kmol/h) and condensing water releases about 40 MJ/kmol. The 17% difference is within what the VLE treatment explains and the 30% area margin in Report 3 (34.8 m^2) would absorb it.

Distillation condenser and reboiler

The hand condenser duty in Report 3 (1,195,877 kJ/h) equals the distillate flow of 35.5 kmol/h multiplied by its latent heat. That is the duty with no reflux. A real column returns reflux, so the condenser must condense (R + 1) times the distillate. At the shortcut column's reflux ratio of 1.5 that alone multiplies the hand duty by 2.5; the larger simulated distillate (40.1 kmol/h) and the two-phase feed account for the rest. The hand condenser and reboiler designs should be revisited with a stated reflux ratio.

Product cooling and solidification

DWSIM recognised the pure benzoic acid product as solid: stream 7 leaves CL-3 at its melting point (122.4°C) as 99.8% solid, with 84.5 kW removed. This is almost entirely the latent heat of fusion (about 18 kJ/mol × 15.6 kmol/h ≈ 78 kW). The hand value of 138 kW also includes sensible cooling to 25 °C, which the simulation does not continue to, so the two are consistent in size. In the hand crystallisation calculation in Report 1, the 18,000 kJ fusion term appears to be added once rather than per kmol, which is worth checking. The simulated bottoms cooler (CL-2) is not comparable, because the separator delivers the bottoms at 125°C rather than 250°C, so CL-2 adds 4.0 kW of heat instead of removing 94 kW.

6. Production rate and scale-up

Quantity	Hand (Report 1)	DWSIM
Benzoic acid product (kmol/h)	14.33	15.59
Product mass flow (kg/h)	1,748	1,904
Annual output at 8,000 h/yr (t/yr)	13,986	15,234
Scale factor for 12,000 t/yr	0.858	0.788
Scaled air feed (kmol/h)	107.3 (Report 2)	98.5
Scaled fresh toluene (kmol/h)	–	13.4

Table 6. Production rate and scale-up.

At the hand basis the simulated plant makes 15,234 t/yr, 9% more than the hand estimate, because the recycle is fully converged. Meeting 12,000 t/yr therefore needs only 0.788 of the base-case feed, against the 0.858 used for

the reactor design. The 18.6 m^3 reactor in Report 2 was sized on the higher flow, so it carries about 9% extra capacity on top of its design margin. The off-gas O_2/N_2 ratio is independent of scale and the simulated 0.029 sits well inside the 0.0625 limit used for flammability safety.

7. Limitations

- **Ideal VLE.** Raoult's law treats toluene and water as miscible outside the decanter. An activity-coefficient model such as NRTL would represent their immiscibility and improve the vent and flash-drum results.
- **Specified separation.** Product purity (100%) and benzoic acid recovery (99%) are inputs, so the simulation cannot test the hand result of 97.5% purity, which is below the 98% minimum in the design brief.
- **Column duties.** The condenser and reboiler duties come from the shortcut column on the once-through flowsheet, not the final recycle case.
- **Fixed conversion.** The 15% conversion is imposed, as in the hand work; the kinetic reactor model in Report 2 is not used.
- **Recycle tolerance.** The recycle converged to 0.17% average error, which causes the 0.69% mass-balance gap.

8. Conclusions and recommendations

The simulation confirms the group's process structure and material balance: the reactor, vent and flash-drum flows agree within a few percent, and the plant meets the 12,000 tonne/yr target with margin. It also shows that the hand energy balance and water balance need revision before the equipment designs are final.

- **Recalculate the reactor energy balance** using a single inlet state for each feed stream; expect a cooling duty near 1.4–1.5 MW rather than 3.6 MW, then resize the jacket and coil.
- **Close the water balance** by routing all distillate through the decanter, so about 15 kmol/h of water leaves there; resize the decanter for this flow.
- **Redo the column duties** with a stated reflux ratio; the current hand values correspond to zero reflux.
- **Check product purity:** the hand bottoms purity of 97.5% is below the 98% specification.
- **Use the 0.788 scale factor** from the converged recycle for final equipment sizing.

Appendix: stream table

All streams from the converged simulation. Flows in kmol/h unless stated.

Stream	T (°C)	P (bar)	Vap. frac.	Toluene	O_2	N_2	Benzoic acid	Water	Total	Mass (kg/h)
1	25.0	8.00	0.884	17.00	26.25	98.75	0	0	142.00	5,172.6
21	34.8	8.00	0.549	103.95	26.25	98.75	0.54	0	229.49	13,249.9
2	165.0	8.00	1.000	52.70	2.86	98.75	0.38	11.27	165.96	7,963.1
9	165.0	8.00	0	35.66	~0	~0	15.75	4.33	55.73	5,286.8
12	45.0	8.00	0.620	52.70	2.86	98.75	0.38	11.27	165.96	7,963.1
15	45.0	8.00	1.000	1.02	2.86	98.75	~0	0.22	102.84	2,955.5
8	45.0	8.00	0	51.68	~0	~0	0.38	11.05	63.12	5,007.6
14	45.4	1.01	~0	51.68	~0	~0	0.38	11.05	63.12	5,007.6
16	108.7	1.01	0.225	87.34	~0	~0	0.54	15.38	103.26	8,390.2
4	40.0	1.01	0	~0	~0	~0	0	15.38	15.38	277.0
17	40.0	1.01	0	87.34	~0	~0	0.54	0	87.88	8,113.0
18	40.0	1.01	1.000	~0	~0	~0	~0	~0	~0	0.2
19	40.0	1.01	0	87.34	~0	~0	0.54	0	87.88	8,113.0
20	40.2	8.00	0	86.95	~0	~0	0.54	0	87.49	8,077.3
13	125.3	1.01	0.244	35.66	~0	~0	15.75	4.33	55.73	5,286.8
5	125.3	1.01	0.998	35.66	~0	~0	0.16	4.33	40.14	3,382.6
11	125.3	1.01	0	0	0	0	15.59	0	15.59	1,904.3
6	130.0	1.01	0	0	0	0	15.59	0	15.59	1,904.3
7	122.4	1.01	0	0	0	0	15.59	0	15.59	1,904.3